

Coach Kai HeyGen Interface Fix - Root Cause Analysis

Date: December 28, 2025

Issue: HeyGen video interface still showing on live site despite previous fix attempts

Status:  RESOLVED

The Real Problem

The HeyGen interface wasn't showing due to the code being wrong - **it was showing because the Vercel build was failing**. When a build fails, Vercel continues serving the last successful build, which still contained the old HeyGen interface.

Evidence

- Vercel deployment logs showed TypeScript compilation errors
 - Build was failing at the type checking stage
 - The coach page (`app/train/coach/page.tsx`) was already correctly importing `SimpleCoachKai`
 - The `SimpleCoachKai` component was properly implemented with text chat and push-to-talk
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Root Cause: TypeScript Build Error

File: `app/api/training/mark-day-complete/route.ts`

Line: 76

The Error

```
// BEFORE (BROKEN):
const completedDaysArray = Array.isArray(userProgram.completedDays)
  ? userProgram.completedDays.map((date: any) => new Date(date))
  : []

let updatedCompletedDays: Date[] = [...completedDaysArray]
//                                     ^^^ Type 'JsonValue[]' is not assignable to
type 'Date[]'
```

The Fix

```
// AFTER (FIXED):
const completedDaysArray: Date[] = Array.isArray(userProgram.completedDays)
  ? userProgram.completedDays.map((date: any) => new Date(date))
  : []
//                                     ^^^^^^ Explicit type annotation resolves the spread operator
type inference issue

let updatedCompletedDays: Date[] = [...completedDaysArray]
```

Explanation: TypeScript couldn't infer that `completedDaysArray` was `Date[]` because Prisma returns `Json` types as `JsonValue`. The spread operator then failed because it saw `JsonValue[]` instead of `Date[]`. Adding an explicit type annotation forces TypeScript to treat the mapped result as `Date[]`.

Missing Dependencies Fixed

The build also failed due to missing npm packages that were imported but not installed:

```
npm install react-dropzone canvas-confetti nodemailer @types/nodemailer
```

Why These Were Missing

- `react-dropzone` : Used in video upload components (`admin-video-upload.tsx` , `video-analysis-hub.tsx`)
- `canvas-confetti` : Used in `day-completion-modal.tsx` for celebration animations
- `nodemailer` : Used in email APIs (`auth/request-reset` , `rewards/redeem`)
- `@types/nodemailer` : TypeScript definitions for nodemailer

Bonus Fix: Broken YouTube Video in Drill Library

Issue: "Angle Serves That Win Points" video was showing as unavailable

Broken Video ID: `tnyUYMjmtzM`

Fix: Replaced with working video `BmdnJNCEwxI`

Changes to `lib/drills-data.ts`

```
// drill: "Short Angle Serve" (serve-008)
videoDemos: [{
  title: "Angle Serves That Win Points",
  url: "https://www.youtube.com/watch?v=BmdnJNCEwxI", // ✅ NEW: Working video
  duration: "8 min",
  description: "Creating angles on serve to pull opponents off court",
  skillLevel: "Intermediate",
  channel: "Pickleball Tutorial"
}]
```

Verification Steps Completed

1. ✅ Confirmed coach page imports `SimpleCoachKai` (not HeyGen component)
2. ✅ Verified `SimpleCoachKai` exists and has no HeyGen dependencies
3. ✅ Fixed TypeScript build error in `mark-day-complete` route
4. ✅ Installed all missing dependencies
5. ✅ Successfully built locally with `npm run build`
6. ✅ Replaced broken YouTube video in drill library
7. ✅ Committed and pushed all changes to GitHub

Deployment Status

Commits Pushed:

1. `40218cf` - "CRITICAL FIX: Restore Coach Kai chat interface"
2. `d48992e` - "Fix broken YouTube video in drill library"

Expected Result:

- Vercel will detect the new commits and trigger a fresh deployment
 - The build will now succeed (TypeScript errors resolved)
 - Once deployed, the live site will show **SimpleCoachKai** text chat interface
 - The broken drill video will now play correctly
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What Users Will See Now

Coach Kai Interface

-  **No more HeyGen video avatar** (which was stuck on "Starting up...")
-  **Beta banner** explaining video is temporarily unavailable
-  **Large "K" avatar** with animated pulse when thinking
-  **Text chat interface** with message history
-  **Push-to-talk button** using browser speech recognition
-  **Quick prompt buttons** (Serve Tips, Dinking, Third Shot, Mental Game)
-  **Mobile-optimized** responsive layout

Drill Library

-  **"Short Angle Serve" video** now plays correctly
 -  **No more "Video unavailable"** errors for this drill
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Technical Details for Future Reference

Why Previous "Fixes" Didn't Work

The code was already correct in the repository. The issue was that:

1. The build was failing silently in Vercel
2. Vercel kept serving the old (cached) successful build
3. Even though we "fixed" the code multiple times, the build never succeeded to deploy it
4. The root cause (TypeScript error) was hidden in build logs, not the application code

How to Prevent This in the Future

1. **Always check Vercel build logs** when changes don't appear on live site
 2. **Test builds locally** before pushing: `npm run build`
 3. **Monitor TypeScript errors** during development
 4. **Keep dependencies in sync** with imports in code
 5. **Use explicit type annotations** when working with Prisma Json types
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Next Steps for User

1. **Check Vercel Dashboard** - Watch for new deployment triggered by recent commits
 2. **Wait 3-5 minutes** for build and deployment to complete
 3. **Clear browser cache** or open in incognito mode
 4. **Visit** `https://mindfulchampion.com/train/coach`
 5. **Verify** text chat interface appears (not HeyGen video)
 6. **Test** the broken drill video at `/train/drills` > "Short Angle Serve"
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Summary

Root Issue: TypeScript build failure due to type inference error with Prisma Json types

Secondary Issues: Missing npm dependencies, broken YouTube video

Resolution: Fixed type annotation, installed dependencies, replaced video

Impact: Build now succeeds, SimpleCoachKai will deploy, drill videos work

Deployment: Automatic via GitHub push to Vercel

Lesson Learned: When "fixes" don't appear on live site, always check the build logs first! 🎯